

Newman-Penrose Formalism Calculations

Stephani et al., Equation (17.4_e-1)

Metric

$$ds^2 = -e^M dt \otimes dt + We^\Psi dx \otimes dx + AWe^\Psi dx \otimes dy + AWe^\Psi dy \otimes dx + \left(A^2 e^{(2\Psi)} + 1\right) We^{(-\Psi)} dy \otimes dy + e^M dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Metric Functions

$$\Psi(t, z) = \Psi$$

$$A(t, z) = A$$

$$W(t, z) = W$$

$$M(t, z) = M$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} e^{(\frac{1}{2} M)}\right) dt + \left(\frac{1}{2} \sqrt{2} e^{(\frac{1}{2} M)}\right) dz$$

$$n = \left(\frac{1}{2} \sqrt{2} e^{(\frac{1}{2} M)}\right) dt + \left(-\frac{1}{2} \sqrt{2} e^{(\frac{1}{2} M)}\right) dz$$

$$m = \left(\frac{1}{2} \sqrt{2} \sqrt{W} e^{\left(\frac{1}{2} \Psi\right)} \right) dx + \left(\frac{1}{2} \sqrt{2} A \sqrt{W} e^{\left(\frac{1}{2} \Psi\right)} + \frac{1}{2} i \sqrt{2} \sqrt{W} e^{\left(-\frac{1}{2} \Psi\right)} \right) dy$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} \sqrt{W} e^{\left(\frac{1}{2} \Psi\right)} \right) dx + \left(\frac{1}{2} \sqrt{2} A \sqrt{W} e^{\left(\frac{1}{2} \Psi\right)} - \frac{1}{2} i \sqrt{2} \sqrt{W} e^{\left(-\frac{1}{2} \Psi\right)} \right) dy$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} e^{\left(-\frac{1}{2} M\right)} \right) dt + \left(\frac{1}{2} \sqrt{2} e^{\left(-\frac{1}{2} M\right)} \right) dz$$

$$n = \left(-\frac{1}{2} \sqrt{2} e^{\left(-\frac{1}{2} M\right)} \right) dt + \left(-\frac{1}{2} \sqrt{2} e^{\left(-\frac{1}{2} M\right)} \right) dz$$

$$m = \left(-\frac{i \sqrt{2} A e^{\left(\frac{1}{2} \Psi\right)}}{2 \sqrt{W}} + \frac{\sqrt{2} e^{\left(-\frac{1}{2} \Psi\right)}}{2 \sqrt{W}} \right) dx + \left(\frac{i \sqrt{2} e^{\left(\frac{1}{2} \Psi\right)}}{2 \sqrt{W}} \right) dy$$

$$\bar{m} = \left(\frac{i \sqrt{2} A e^{\left(\frac{1}{2} \Psi\right)}}{2 \sqrt{W}} + \frac{\sqrt{2} e^{\left(-\frac{1}{2} \Psi\right)}}{2 \sqrt{W}} \right) dx + \left(-\frac{i \sqrt{2} e^{\left(\frac{1}{2} \Psi\right)}}{2 \sqrt{W}} \right) dy$$

Directional Derivatives

$$D = -\frac{1}{2} \sqrt{2} e^{\left(-\frac{1}{2} M\right)} \partial_t + \frac{1}{2} \sqrt{2} e^{\left(-\frac{1}{2} M\right)} \partial_z$$

$$\Delta = -\frac{1}{2} \sqrt{2} e^{\left(-\frac{1}{2} M\right)} \partial_t - \frac{1}{2} \sqrt{2} e^{\left(-\frac{1}{2} M\right)} \partial_z$$

$$\delta = -\frac{i \sqrt{2} A e^{\left(\frac{1}{2} \Psi\right)}}{2 \sqrt{W}} + \frac{\sqrt{2} e^{\left(-\frac{1}{2} \Psi\right)}}{2 \sqrt{W}} \partial_x + \frac{i \sqrt{2} e^{\left(\frac{1}{2} \Psi\right)}}{2 \sqrt{W}} \partial_y$$

$$\bar{\delta} = \frac{i\sqrt{2}Ae^{(\frac{1}{2}\Psi)}}{2\sqrt{W}} + \frac{\sqrt{2}e^{(-\frac{1}{2}\Psi)}}{2\sqrt{W}}\partial_x - \frac{i\sqrt{2}e^{(\frac{1}{2}\Psi)}}{2\sqrt{W}}\partial_y$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial t}W}{4W} - \frac{\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial z}W}{4W}$$

$$\sigma = \frac{1}{4}i\sqrt{2}e^{(-\frac{1}{2}M+\Psi)}\frac{\partial}{\partial t}A - \frac{1}{4}i\sqrt{2}e^{(-\frac{1}{2}M+\Psi)}\frac{\partial}{\partial z}A + \frac{1}{4}\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial t}\Psi - \frac{1}{4}\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial z}\Psi$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial t}W}{4W} - \frac{\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial z}W}{4W}$$

$$\lambda = \frac{1}{4}i\sqrt{2}e^{(-\frac{1}{2}M+\Psi)}\frac{\partial}{\partial t}A + \frac{1}{4}i\sqrt{2}e^{(-\frac{1}{2}M+\Psi)}\frac{\partial}{\partial z}A - \frac{1}{4}\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial t}\Psi - \frac{1}{4}\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial z}\Psi$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{1}{8}i\sqrt{2}e^{(-\frac{1}{2}M+\Psi)}\frac{\partial}{\partial t}A - \frac{1}{8}i\sqrt{2}e^{(-\frac{1}{2}M+\Psi)}\frac{\partial}{\partial z}A - \frac{1}{8}\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial t}M + \frac{1}{8}\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial z}M$$

$$\gamma = \frac{1}{8} i \sqrt{2} e^{(-\frac{1}{2} M + \Psi)} \frac{\partial}{\partial t} A + \frac{1}{8} i \sqrt{2} e^{(-\frac{1}{2} M + \Psi)} \frac{\partial}{\partial z} A + \frac{1}{8} \sqrt{2} e^{(-\frac{1}{2} M)} \frac{\partial}{\partial t} M + \frac{1}{8} \sqrt{2} e^{(-\frac{1}{2} M)} \frac{\partial}{\partial z} M$$

Ricci Tensor Components

$$\begin{aligned} \Phi_{00} = & -\frac{1}{8} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A^2 + \frac{1}{4} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial z} A - \frac{1}{8} e^{(-M+2\Psi)} \frac{\partial}{\partial z} A^2 - \frac{1}{8} e^{(-M)} \frac{\partial}{\partial t} \Psi^2 + \frac{1}{4} e^{(-M)} \frac{\partial}{\partial t} \Psi \frac{\partial}{\partial z} \Psi - \frac{1}{8} e^{(-M)} \frac{\partial}{\partial z} \Psi^2 + \frac{e^{(-M)} \frac{\partial}{\partial t} M \frac{\partial}{\partial t} W}{4W} - \frac{e^{(-M)} \frac{\partial}{\partial z} M \frac{\partial}{\partial t} W}{4W} \\ & - \frac{e^{(-M)} \frac{\partial}{\partial t} M \frac{\partial}{\partial z} W}{4W} + \frac{e^{(-M)} \frac{\partial}{\partial z} M \frac{\partial}{\partial z} W}{4W} + \frac{e^{(-M)} \frac{\partial}{\partial t} W^2}{8W^2} - \frac{e^{(-M)} \frac{\partial^2}{(\partial t)^2} W}{4W} + \frac{e^{(-M)} \frac{\partial^2}{\partial t \partial z} W}{2W} - \frac{e^{(-M)} \frac{\partial}{\partial t} W \frac{\partial}{\partial z} W}{4W^2} + \frac{e^{(-M)} \frac{\partial}{\partial z} W^2}{8W^2} - \frac{e^{(-M)} \frac{\partial^2}{(\partial z)^2} W}{4W} \end{aligned}$$

$$\Phi_{01} = 0$$

$$\begin{aligned} \Phi_{02} = & -\frac{1}{4} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A^2 + \frac{1}{4} e^{(-M+2\Psi)} \frac{\partial}{\partial z} A^2 + \frac{1}{2} i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} \Psi - \frac{1}{2} i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial z} \Psi + \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial^2}{(\partial t)^2} A - \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial^2}{(\partial z)^2} A \\ & + \frac{1}{4} e^{(-M)} \frac{\partial^2}{(\partial t)^2} \Psi - \frac{1}{4} e^{(-M)} \frac{\partial^2}{(\partial z)^2} \Psi + \frac{i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} W}{4W} + \frac{e^{(-M)} \frac{\partial}{\partial t} \Psi \frac{\partial}{\partial t} W}{4W} - \frac{i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial z} W}{4W} - \frac{e^{(-M)} \frac{\partial}{\partial z} \Psi \frac{\partial}{\partial z} W}{4W} \end{aligned}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{1}{16} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A^2 + \frac{1}{16} e^{(-M+2\Psi)} \frac{\partial}{\partial z} A^2 - \frac{1}{16} e^{(-M)} \frac{\partial}{\partial t} \Psi^2 + \frac{1}{16} e^{(-M)} \frac{\partial}{\partial z} \Psi^2 - \frac{1}{8} e^{(-M)} \frac{\partial^2}{(\partial t)^2} M + \frac{1}{8} e^{(-M)} \frac{\partial^2}{(\partial z)^2} M + \frac{e^{(-M)} \frac{\partial}{\partial t} W^2}{16W^2} - \frac{e^{(-M)} \frac{\partial}{\partial z} W^2}{16W^2}$$

$$\Phi_{12} = 0$$

$$\begin{aligned} \Phi_{20} = & -\frac{1}{4} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A^2 + \frac{1}{4} e^{(-M+2\Psi)} \frac{\partial}{\partial z} A^2 - \frac{1}{2} i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} \Psi + \frac{1}{2} i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial z} \Psi - \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial^2}{(\partial t)^2} A + \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial^2}{(\partial z)^2} A \\ & + \frac{1}{4} e^{(-M)} \frac{\partial^2}{(\partial t)^2} \Psi - \frac{1}{4} e^{(-M)} \frac{\partial^2}{(\partial z)^2} \Psi - \frac{i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} W}{4W} + \frac{e^{(-M)} \frac{\partial}{\partial t} \Psi \frac{\partial}{\partial t} W}{4W} + \frac{i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial z} W}{4W} - \frac{e^{(-M)} \frac{\partial}{\partial z} \Psi \frac{\partial}{\partial z} W}{4W} \end{aligned}$$

$$\Phi_{21} = 0$$

$$\begin{aligned}\Phi_{22} = & -\frac{1}{8} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A^2 - \frac{1}{4} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial z} A - \frac{1}{8} e^{(-M+2\Psi)} \frac{\partial}{\partial z} A^2 - \frac{1}{8} e^{(-M)} \frac{\partial}{\partial t} \Psi^2 - \frac{1}{4} e^{(-M)} \frac{\partial}{\partial t} \Psi \frac{\partial}{\partial z} \Psi - \frac{1}{8} e^{(-M)} \frac{\partial}{\partial z} \Psi^2 + \frac{e^{(-M)} \frac{\partial}{\partial t} M \frac{\partial}{\partial t} W}{4W} + \frac{e^{(-M)} \frac{\partial}{\partial z} M \frac{\partial}{\partial t} W}{4W} \\ & + \frac{e^{(-M)} \frac{\partial}{\partial t} M \frac{\partial}{\partial z} W}{4W} + \frac{e^{(-M)} \frac{\partial}{\partial z} M \frac{\partial}{\partial z} W}{4W} + \frac{e^{(-M)} \frac{\partial}{\partial t} W^2}{8W^2} - \frac{e^{(-M)} \frac{\partial^2}{(\partial t)^2} W}{4W} - \frac{e^{(-M)} \frac{\partial^2}{\partial t \partial z} W}{2W} + \frac{e^{(-M)} \frac{\partial}{\partial t} W \frac{\partial}{\partial z} W}{4W^2} + \frac{e^{(-M)} \frac{\partial}{\partial z} W^2}{8W^2} - \frac{e^{(-M)} \frac{\partial^2}{(\partial z)^2} W}{4W}\end{aligned}$$

$$\begin{aligned}\Lambda = & \frac{1}{48} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A^2 - \frac{1}{48} e^{(-M+2\Psi)} \frac{\partial}{\partial z} A^2 + \frac{1}{48} e^{(-M)} \frac{\partial}{\partial t} \Psi^2 - \frac{1}{48} e^{(-M)} \frac{\partial}{\partial z} \Psi^2 + \frac{1}{24} e^{(-M)} \frac{\partial^2}{(\partial t)^2} M \\ & - \frac{1}{24} e^{(-M)} \frac{\partial^2}{(\partial z)^2} M - \frac{e^{(-M)} \frac{\partial}{\partial t} W^2}{48W^2} + \frac{e^{(-M)} \frac{\partial^2}{(\partial t)^2} W}{12W} + \frac{e^{(-M)} \frac{\partial}{\partial z} W^2}{48W^2} - \frac{e^{(-M)} \frac{\partial^2}{(\partial z)^2} W}{12W}\end{aligned}$$

Weyl Tensor Components

$$\begin{aligned}\Psi_0 = & \frac{1}{4} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A^2 - \frac{1}{2} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial z} A + \frac{1}{4} e^{(-M+2\Psi)} \frac{\partial}{\partial z} A^2 + \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} M - \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial t} M - \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial z} M + \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial z} M \\ & - \frac{1}{2} i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} \Psi + \frac{1}{2} i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial t} \Psi + \frac{1}{4} e^{(-M)} \frac{\partial}{\partial t} M \frac{\partial}{\partial t} \Psi - \frac{1}{4} e^{(-M)} \frac{\partial}{\partial z} M \frac{\partial}{\partial t} \Psi + \frac{1}{2} i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial z} \Psi - \frac{1}{2} i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial z} \Psi - \frac{1}{4} e^{(-M)} \frac{\partial}{\partial t} M \frac{\partial}{\partial z} \Psi \\ & + \frac{1}{4} e^{(-M)} \frac{\partial}{\partial z} M \frac{\partial}{\partial z} \Psi - \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial^2}{(\partial t)^2} A + \frac{1}{2} i e^{(-M+\Psi)} \frac{\partial^2}{\partial t \partial z} A - \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial^2}{(\partial z)^2} A - \frac{1}{4} e^{(-M)} \frac{\partial^2}{(\partial t)^2} \Psi + \frac{1}{2} e^{(-M)} \frac{\partial^2}{\partial t \partial z} \Psi - \frac{1}{4} e^{(-M)} \frac{\partial^2}{(\partial z)^2} \Psi - \frac{i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} W}{4W} \\ & + \frac{i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial t} W}{4W} - \frac{e^{(-M)} \frac{\partial}{\partial t} \Psi \frac{\partial}{\partial t} W}{4W} + \frac{e^{(-M)} \frac{\partial}{\partial z} \Psi \frac{\partial}{\partial t} W}{4W} + \frac{i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial z} W}{4W} - \frac{i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial z} W}{4W} + \frac{e^{(-M)} \frac{\partial}{\partial t} \Psi \frac{\partial}{\partial z} W}{4W} - \frac{e^{(-M)} \frac{\partial}{\partial z} \Psi \frac{\partial}{\partial z} W}{4W}\end{aligned}$$

$$\Psi_1 = 0$$

$$\begin{aligned}\Psi_2 = & \frac{1}{12} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A^2 - \frac{1}{12} e^{(-M+2\Psi)} \frac{\partial}{\partial z} A^2 - \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial t} \Psi + \frac{1}{12} e^{(-M)} \frac{\partial}{\partial t} \Psi^2 + \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial z} \Psi - \frac{1}{12} e^{(-M)} \frac{\partial}{\partial z} \Psi^2 \\ & - \frac{1}{12} e^{(-M)} \frac{\partial^2}{(\partial t)^2} M + \frac{1}{12} e^{(-M)} \frac{\partial^2}{(\partial z)^2} M - \frac{e^{(-M)} \frac{\partial}{\partial t} W^2}{12W^2} + \frac{e^{(-M)} \frac{\partial^2}{(\partial t)^2} W}{12W} + \frac{e^{(-M)} \frac{\partial}{\partial z} W^2}{12W^2} - \frac{e^{(-M)} \frac{\partial^2}{(\partial z)^2} W}{12W}\end{aligned}$$

$$\Psi_3 = 0$$

$$\begin{aligned}
\Psi_4 = & \frac{1}{4} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A^2 + \frac{1}{2} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial z} A + \frac{1}{4} e^{(-M+2\Psi)} \frac{\partial}{\partial z} A^2 - \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} M - \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial t} M - \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial z} M - \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial z} M \\
& + \frac{1}{2} i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} \Psi + \frac{1}{2} i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial t} \Psi + \frac{1}{4} e^{(-M)} \frac{\partial}{\partial t} M \frac{\partial}{\partial t} \Psi + \frac{1}{4} e^{(-M)} \frac{\partial}{\partial z} M \frac{\partial}{\partial t} \Psi + \frac{1}{2} i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial z} \Psi + \frac{1}{2} i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial z} \Psi + \frac{1}{4} e^{(-M)} \frac{\partial}{\partial t} M \frac{\partial}{\partial z} \Psi \\
& + \frac{1}{4} e^{(-M)} \frac{\partial}{\partial z} M \frac{\partial}{\partial z} \Psi + \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial^2}{(\partial t)^2} A + \frac{1}{2} i e^{(-M+\Psi)} \frac{\partial^2}{\partial t \partial z} A + \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial^2}{(\partial z)^2} A - \frac{1}{4} e^{(-M)} \frac{\partial^2}{(\partial t)^2} \Psi - \frac{1}{2} e^{(-M)} \frac{\partial^2}{\partial t \partial z} \Psi - \frac{1}{4} e^{(-M)} \frac{\partial^2}{(\partial z)^2} \Psi + \frac{i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} W}{4 W} \\
& + \frac{i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial t} W}{4 W} - \frac{e^{(-M)} \frac{\partial}{\partial t} \Psi \frac{\partial}{\partial t} W}{4 W} - \frac{e^{(-M)} \frac{\partial}{\partial z} \Psi \frac{\partial}{\partial t} W}{4 W} + \frac{i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial z} W}{4 W} + \frac{i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial z} W}{4 W} - \frac{e^{(-M)} \frac{\partial}{\partial t} \Psi \frac{\partial}{\partial z} W}{4 W} - \frac{e^{(-M)} \frac{\partial}{\partial z} \Psi \frac{\partial}{\partial z} W}{4 W}
\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"